Amabwiriza y'umufundi w'amazi: sitidiyo

Ibyo umufundi w'amazi akeneye byose kuri sitidiyo imwe ya kontineri: amazi meza, amazi ashyushye, imiyoboro y'amazi yanduye, amagerageza, n'ibyo sitidiyo ikeneye ku ifamu.

> ***Ubusobanuro:*** *iyi nyandiko yahinduwe na mudasobwa (machine draft). Igomba gusuzumwa n'umuntu uvuga Ikinyarwanda kavukire mbere yo kuyiha abakozi. Iyo hari itandukaniro, kurikiza iy'Icyongereza.*

Status: verisiyo A2, 2026-10-28, PRELIMINARY. Nta mushyitsi ucumbika muri sitidiyo mbere y'uko septic system itsinda igenzura ryayo (D-016).

Amatiyo ya sitidiyo zombi ashyirwa kimwe, bityo aya mabwiriza akora kuri zombi.

English: read this guide in English · Gucapa: aya mabwiriza nka PDF

Amagambo akoreshwa hano

- **End A (impera A)**: urukuta rufunze rw'impera, iruhande rw'igitanda. **End B (impera B)**: impera ifite inzugi za kontineri, iruhande rw'ubwiherero.
- **Back wall (urukuta rw'inyuma)**: urukuta rurerure rudafite aho gufungukira. **Front wall (urukuta rw'imbere)**: urukuta rurerure rufite urugi runyerera rw'ikirahure.
- Imyanya iri mu mbonerahamwe y'ibikoresho (fittings) ibarwa muri milimetero: uhereye ku rukuta rurangiye rwa end A (x), uhereye ku rukuta rw'inyuma rurangiye (y), no kuzamuka uhereye hasi harangiye (z).

Sitidiyo mu ncamake

Iki	Uko byateganyijwe
Itiyo y'amazi meza (supply pipe)	PPR PN20, mm 25 ku muyoboro mukuru (main), mm 20 imbere
Aho umuyoboro mukuru winjirira (main in)	munsi ya hasi kuri end B, ukazamuka muri sink base: stop valve (robine ifunga amazi), Y-filter, check meter (mubazi y'amazi), byose kuri SV
Amazi ashyushye	agasanduku ka water heater ka L 30, 2 kW, IPX4, ku rukuta rwa end B hejuru ya WC; amazi abikwa kuri 60 °C

Iki	Uko byateganyijwe
Valve ivanga amazi (mixing valve, TMV)	inyuma y'urugi ruto rwo kuyigeraho (access panel) rutwikiriwe n'amakaro, mu rukuta rwa end B, hagati ya douche na WC; itanga amazi ya 41 kugeza 44 °C kuri douche no kuri lavabo
Imiyoboro y'amazi yanduye (drains)	WC mm 110, douche mm 50, lavabo mm 40, sink y'igikoni mm 50 ku muyoboro wayo wihariye ujya kuri grease trap (ifata amavuta); yose ifite ubuhaname bwa 1:40 cyangwa burenzeho
Umuyoboro w'umwuka (vent)	open vent ya mm 75 izamuka kuri end B, ikarangirira mm 900 hejuru y'uruzitiro rwo ku gisenge (guardrail)
Tray yo kwiyuhagiriramo (shower tray)	1000 × 900
Umuvuduko w'amazi (pressure)	bar 2.5 kugeza 3, uturuka kuri pompe yongera umuvuduko (booster pump) y'ifamu

Ryari	Iki	Intambwe
Nyuma y'ibiti (battens, S03)	First fix: umuyoboro mukuru winjira, buri muyoboro w'amazi meza n'uw'amazi yanduye, igerageza rya bar 10	S04
Hamwe n'ubwiherero	Tray (S09), WC, heater na lavabo (S10)	S09, S10
Hamwe n'igikoni	Sink, robine n'akayunguruzo (filter)	S13

Ingingo eshatu zo guhagarara ukagenzura (hold points)

1. **Ibiti mbere.** Tangira first fix gusa iyo ibiti (S03) byamaze gufatwa kandi impapuro za hasi zamaze gushyirwaho.
2. **Igerageza rya bar 10.** Uzuza amazi mu matiyo ya PPR, ugumane bar 10 mu gihe cy'isaha imwe nta kugabanuka, ubyandike kandi ubisinyire mbere ya spray foam (S05). Foam ihisha amatiyo burundu.
3. **Mbere y'abashyitsi.** Imiyoboro y'amazi yanduye yageragejwe yuzuzwa amazi (fill-test), vent yageragejwe n'umwotsi (smoke-test), TMV yapimwe kuri douche, kandi septic system y'ifamu yagenzuwe.

S04: first fix

1. Zana umuyoboro mukuru w'amazi akonje (W1, mm 25) unyuze munsi ya hasi kuri end B, uwuzamure muri sink base unyuze kuri stop valve, Y-filter na check meter, aho amazi ashobora gufungwa nta gikoresho.

2. Nyuza amazi akonje imbere mu tubati no mu nkuta: unyuze muri partition, muri lining y'urukuta rw'inyuma no mu rukuta rwa end B, ugere kuri cistern ya WC (W4), kuri lavabo (W5) no hejuru kuri heater (W2). Amatiyo y'amazi akonje agume mm 300 hejuru ya hasi.
3. Manura amazi ashyushye (60 °C) uvuye kuri heater ugere kuri mixing valve (H1), hanyuma uyagarure ukurikiye izo nkuta zimwe ugere kuri sink (H2, yanditseho ko ari amazi ashyushye). Amatiyo y'amazi ashyushye agume mm 360 hejuru ya hasi.
4. Uhereye kuri mixing valve, shyiraho imiyoboro ibiri y'amazi avanze (tempered runs): T1 ijya kuri valve ya douche, T2 ijya kuri lavabo. Buri muyoboro ni hafi m 1.2, munsi ya m 2 zemewe ku itiyo y'amazi avanze ahagararamo (dead leg).
5. Buri muyoboro ugenda ugororotse: ukurikiye urukuta, hanyuma ukazamuka ugororotse ugera aho usohokera (outlet). Nta gice cy'itiyo kigenda kiberamye (diagonal).
6. Manika imiyoboro y'amazi yanduye mu mwanya uri munsi ya hasi (kontineri ihagaze hafi mm 700 hejuru y'ubutaka): WC mm 110, douche mm 50, lavabo mm 40, ijya kuri inspection chamber (regard); sink y'igikoni mm 50 ku muyoboro wayo wihariye ujya kuri grease trap. Nyura hagati y'ibyuma byambukiranya (cross-members) bya mm 122, ntuzigere ubinyuramo.
7. Ubuhaname (fall) bwa 1:40 cyangwa burenzeho; rodding eye (aho basukurira umuyoboro) kuri buri coude (bend); vent ya mm 75 izamuka kuri end B.
8. Kora igerageza ry'umuvuduko kuri bar 10 mu gihe cy'isaha imwe, ufotore buri rukuta, usinye uruhushya rwo gushyiraho foam (release).

Imiyoboro: ubwoko, ingano, uburebure n'ubuhaname · download CSV

run	system	dn	length_m	fall	note	colour
D1	soil	110	1.8	1:38	WC to IC1, 1:40	#8b4a12
D2	waste	50	1.3	1:34	Shower (90 mm tray waste) joins the soil run	#8b4a12
D3	waste	40	2.1	1:28	Basin (wash lobby) joins the soil run	#8b4a12
D4	waste	50	3.3	1:26	Kitchen sink to the grease trap (separate soak pit)	#8b4a12
V1	vent	75	6.7		Open vent up end B to 900 above the guardrail	#9a9893
W1	cold	25	2.9		Cold main in from the farm, up into the sink base: stop valve, filter, check meter	#2a78d6
W2	cold	20	1.8		Cold up end wall B to the heater	#2a78d6
W3	cold	20	0.1		Cold to the sink's angle valve	#2a78d6

run	system	dn	length_m	fall	note	colour
W4	cold	20	5		Cold through the partition and the walls to the WC cistern	#2a78d6
W5	cold	20	1		Cold on to the basin	#2a78d6
W6	cold	20	0.1		Cold to the mixing valve	#2a78d6
H1	hot	20	2.4		Hot (60 C) from the heater down to the mixing valve	#e34948
H2	hot	20	3.9		Hot back along the walls to the sink (direct 60 C, labelled)	#e34948
T1	tempered	20	1.2		TMV 41-44 C up to the shower valve	#eda100
T2	tempered	20	1.2		Tempered to basin	#eda100

Ibikoresho (fittings): icyo ari cyo, aho kiri n'uko kiri hejuru · download CSV

point	kind	model	runs	description	item	x_mm	y_mm
SV	valve	meter	W1	Stop valve, Y-filter and check meter on the main, in the sink base	Stop valve + Y-filter + check water meter	4578	1540
WH	heater	heater	W2 H1 H2	30 L water heater, a box on end wall B above the WC, its T&P valve piped outside	Electric storage water heater 30 L, 2 kW, IPX4, with T&P valve	5768	1300
TMV	valve	tmv	H1 W6 T1 T2	Mixing valve behind an access panel in end wall B, between the shower and the WC	Thermostatic mixing valve (TMV) 41-44 C	5768	1005

point	kind	model	runs	description	item	x_mm	y_mm
KT	tap	kitchentap	W3 H2 D4	Kitchen mixer tap, angle valves under the sink	Kitchen mixer tap	4528	1375
DF	filter	filter	W3	Drinking-water filter in the sink base, its own faucet by the tap	Under-sink drinking-water filter + faucet	4278	1168
SH	shower	showerset	T1 D2	Thermostatic shower valve, head and rail on end wall B	Thermostatic shower valve + 6-7 L/min head + rail	5768	450
WC	wc	cistern	W4 D1	Concealed cistern in end wall B, dual-flush plate	Slim concealed cistern frame (80 mm) with dual-flush plate, in the end-B stud wall and its 30 mm build-out	5768	1300
BT	tap	basintap	W5 T2 D3	Basin mixer, angle valves under the console	Wall basin 400 x 300 + mixer + bottle trap (wash lobby)	5768	2020
IC1	site	chamber	D1 D2 D3	Inspection chamber outside end B, on to the farm's septic tank	Septic tank (2 chambers, about 4 m3) + inspection chambers	6698	1126
GT	site	greasetrap	D4	To the farm's grease trap and its own soak pit	Grease trap + kitchen soak pit	6698	1375

point	kind	model	runs	description	item	x_mm	y_mm
VT	vent	vent	V1	Vent terminal, 900 mm above the roof guardrail	uPVC vent pipe DN75	5918	200

S09, S10 na S13: second fix

1. S09: shyiraho tray yo kwiyuhagiriramo (1000 × 900) ku mwobo wayo usohora amazi (waste) wa mm 90, hamwe na ledge y'umubaji n'ikirahure gihamye (screen).
2. S10: manika WC ifatwa ku rukuta (wall-hung) kuri cistern yayo yoroheje ihishe mu rukuta (concealed cistern): ikadiri ya mm 80 mu rukuta rwa end B, urukuta rwongerewe mm 30 imbere (build-out), na dual-flush plate.
3. S10: manika agasanduku ka heater ka L 30 (450 × 450 × 300) hejuru ya WC; uyobore hanze itiyo ya valve yayo irekura ubushyuhe n'umuvuduko birenze (temperature-and-pressure relief valve).
4. S10: shyiraho lavabo (400 × 300) na mixer yayo hamwe na bottle trap (siphon) mu gice cyo gukarabiramo (wash lobby); shyiraho na valve ya douche igenga ubushyuhe (thermostatic shower valve), umutwe wa douche (head) na rail ku rukuta rwa end B.
5. S13: robine ivanga y'igikoni (kitchen mixer tap) kuri angle valves, n'akayunguruzo k'amazi yo kunywa (drinking-water filter) muri sink base, gafite robine yako bwite iruhande rwa robine ivanga.

Amagerageza (tests)

1. PPR kuri bar 10 mu gihe cy'isaha imwe, mbere ya foam (S04).
2. Gerageza buri muyoboro w'amazi yanduye uwuzuza amazi (fill-test), kandi ugenzure ko buri siphon (trap) igumana amazi ayifunga (seal).
3. Gerageza vent ukoresheje umwotsi (smoke-test).
4. Pima amazi avanze kuri douche ukoresheje igipimabushyuhe (thermometer): 41 kugeza 44 °C.
5. Fungura douche y'amazi ashyushye, hanyuma ubare igihe heater imara: hafi iminota 40 kugira ngo yongere gushyushya amazi.

Ibyo sitidiyo ikeneye ku ifamu

- Umuyoboro umwe w'amazi akonje ufite bar 2.5 kugeza 3, uvuye ku kigega cy'amazi cy'ifamu (storage tank) no kuri pompe yongera umuvuduko (booster pump). Amazi y'umuyoboro rusange

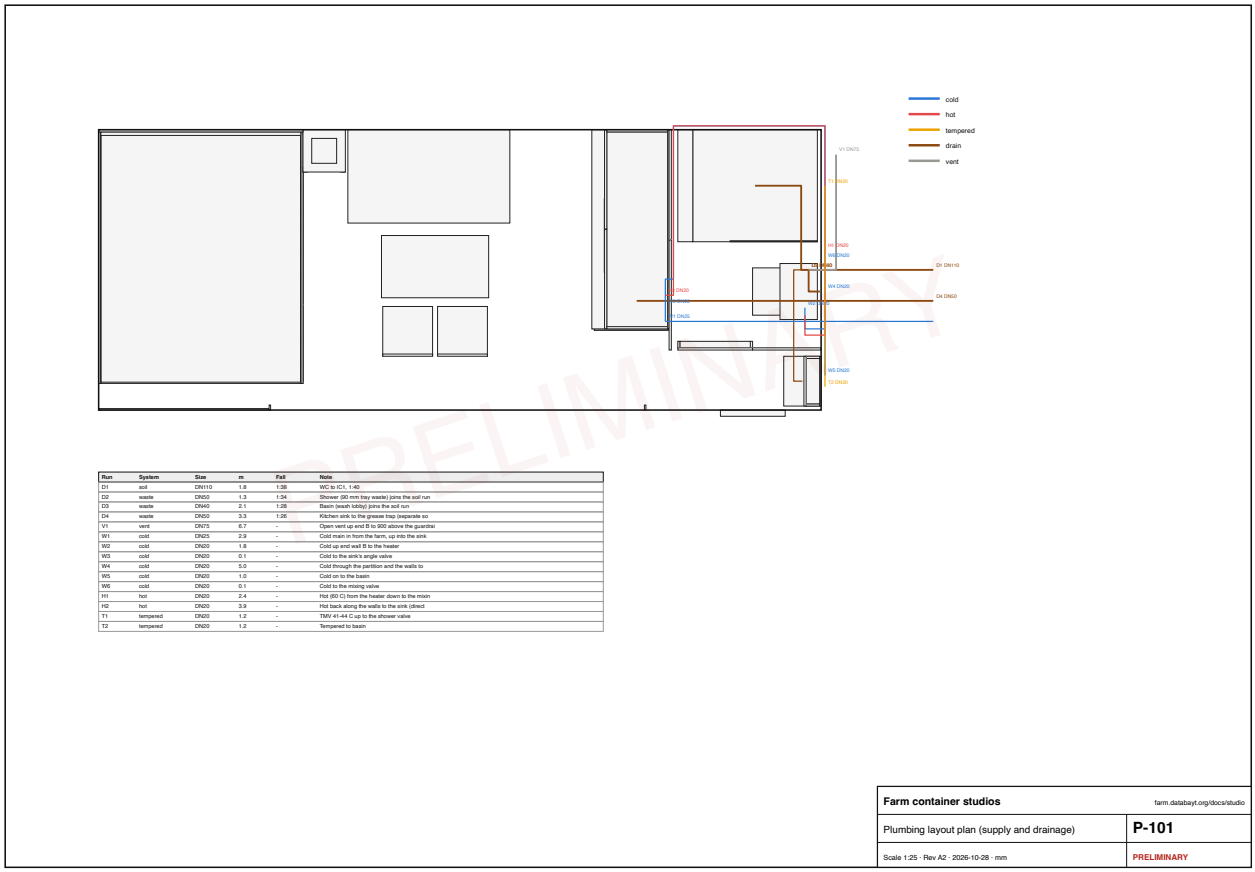
(mains) yinjira muri icyo kigega anyuze ku mwanya w'umwuka (air gap), kugira ngo amazi y'ifamu atazigera asubira mu muyoboro rusange (D-016).

- Umuyoboro w'imyanda (soil connection) uva kuri inspection chamber iri hanze ya end B ujya kuri septic tank (fosse septique) y'ifamu, n'umuyoboro w'igikoni ujya kuri grease trap ifite soak pit (puits perdu) yayo bwite.
- Septic tank ishyirwa nibura m 50 uvuye kuri borehole (iriba ricukuwe), m 3 kugeza 7 uvuye ku nyubako no ku mbibi z'ubutaka (boundary), hepfo ya borehole (downhill), itarenze m 30 uvuye aho imodoka ivoma imyanda (tanker) igera, kandi ishyirwaho nyuma ya percolation test gusa (igerageza ry'uko ubutaka bunywa amazi).

Amazi y'ikibanza n'ingano ya septic tank kuri sitidiyo zombi · download CSV

item	value
persons	4
daily_L	600
tank_m3	2.5
septic_rs143_m3	3
septic_2012_m3	2.72
septic_uk_m3	3.08
septic_m3	3.5
chamber1_m3	2.33
chamber2_m3	1.17
pump_lpm	26

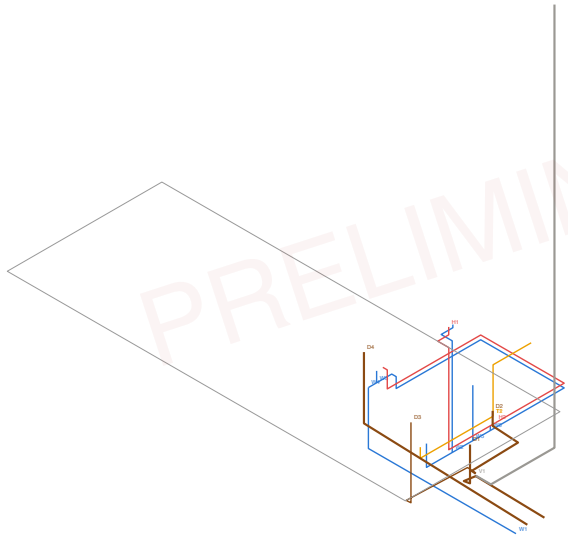
Ibishushanyo



Run	System	Size	m	Fall	Note
D1	soil	DN110	1.8	1:38	WC to IC1, 1:40
D2	waste	DN50	1.3	1:34	Shower (90 mm tray waste) joins the soil run
D3	waste	DN40	2.1	1:28	Basin (wash lobby) joins the soil run
D4	waste	DN50	3.3	1:20	Kitchen sink to the grease trap (separate so
V1	vent	DN75	6.7	-	Open vent up end B to 900 above the guardrai
W1	cold	DN25	2.9	-	Cold main in from the farm, up into the sink
W2	cold	DN20	1.8	-	Cold up end wall B to the heater
W3	cold	DN20	0.1	-	Cold to the sink's angle valve
W4	cold	DN20	5.0	-	Cold through the partition and the walls to
W5	cold	DN20	1.0	-	Cold on to the basin
W6	cold	DN20	0.1	-	Cold to the mixing valve
H1	hot	DN20	2.4	-	Hot (60 C) from the heater down to the mixin
H2	hot	DN20	3.9	-	Hot back along the walls to the sink (direct
T1	tempered	DN20	1.2	-	TMV 41–44 C up to the shower valve
T2	tempered	DN20	1.2	-	Tempered to basin

P-101 Plumbing layout plan · open full size

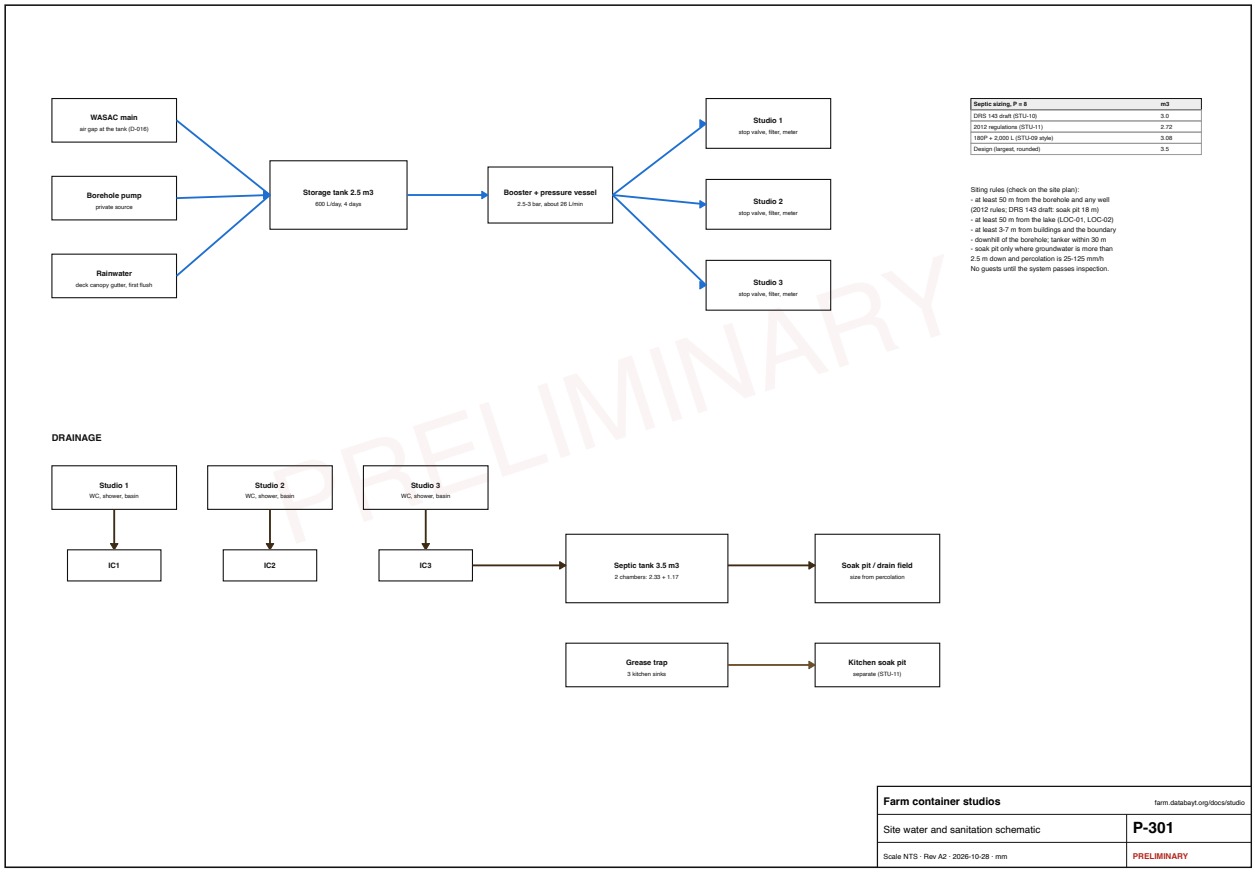
Supply: PPR PN20, 20/25 mm. Hot leaves the 30 L heater at 60 C; a thermostatic mixer (TMV) delivers 41-44 C to the shower and basin; the kitchen sink gets direct hot (labelled).
Drainage hangs in the void under the raised floor (the container is about 700 mm up), between cross-members: WC 110, shower and sink 50, basin 40, falls at least 1:40.
Vent: 75 mm open vent up end B to 900 above the guardrail line (or an AAV inside).
Kitchen sink waste runs separately to a grease trap and its own soak pit (STU-11).



PRELIMINARY

Farm container studios	farm.databayt.org/docs/studio
Pipework isometric (unit)	**P-201**
Scale NTS · Rev A2 · 2026-10-28 · mm	PRELIMINARY

P-201 Pipework isometric · open full size



P-301 Site water and sanitation schematic · open full size